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September 27, 2017
GO2-17-165

10 CFR 50.73

U.S. Nuclear Regulatory Commission
ATTN: Document Control Desk
Washington, D.C. 20555-0001

Subject: **COLUMBIA GENERATING STATION, DOCKET NO. 50-397**
LICENSEE EVENT REPORT NO. 2016-002-01

Dear Sir or Madam:

Transmitted herewith is Licensee Event Report No. 2016-002-01 for Columbia Generating Station, a supplement to the LER submitted on November 17, 2016. This report is submitted pursuant to 10 CFR 50.73(a)(2)(v)(C) and 10 CFR 50.73(a)(2)(v)(D).

There are no commitments being made to the Nuclear Regulatory Commission by this letter. If you have any questions or require additional information, please contact Ms. D.M. Wolfgramm, Regulatory Compliance Supervisor, at (509) 377-4792.

Executed on this 27th day of September, 2017

Respectfully,


A. L. Javorik
Vice President, Engineering

Enclosure: Licensee Event Report 2016-002-01

cc: NRC Region IV Administrator
NRC NRR Project Manager
NRC Senior Resident Inspector/988C
CD Sonoda – BPA/1399
WA Horin – Winston & Strawn

**LICENSEE EVENT REPORT (LER)**

(See Page 2 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the Information Services Branch (T-2 F43), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollects.Resource@nrc.gov, and to the Desk Officer, Office of Information and Regulatory Affairs, NEOF-10202, (3150-0104), Office of Management and Budget, Washington, DC 20503. If a means used to impose an information collection does not display a currently valid OMB control number, the NRC may not conduct or sponsor, and a person is not required to respond to, the information collection.

1. FACILITY NAME

Columbia Generating Station

2. DOCKET NUMBER

05000 397

3. PAGE

1 OF 3

4. TITLE

Valve Closure Results in Momentary Increase in Secondary Containment Pressure

5. EVENT DATE			6. LER NUMBER			7. REPORT DATE			8. OTHER FACILITIES INVOLVED			
MONTH	DAY	YEAR	YEAR	SEQUENTIAL NUMBER	REV NO.	MONTH	DAY	YEAR	FACILITY NAME	DOCKET NUMBER		
10	03	2016	2016	002	01	09	27	2017	FACILITY NAME	DOCKET NUMBER		
										05000		
										05000		
9. OPERATING MODE			11. THIS REPORT IS SUBMITTED PURSUANT TO THE REQUIREMENTS OF 10 CFR §: (Check all that apply)									
1			☐ 20.2201(b)			☐ 20.2203(a)(3)(i)			☐ 50.73(a)(2)(ii)(A)		☐ 50.73(a)(2)(viii)(A)	
			☐ 20.2201(d)			☐ 20.2203(a)(3)(ii)			☐ 50.73(a)(2)(ii)(B)		☐ 50.73(a)(2)(viii)(B)	
			☐ 20.2203(a)(1)			☐ 20.2203(a)(4)			☐ 50.73(a)(2)(iii)		☐ 50.73(a)(2)(ix)(A)	
			☐ 20.2203(a)(2)(i)			☐ 50.36(c)(1)(i)(A)			☐ 50.73(a)(2)(iv)(A)		☐ 50.73(a)(2)(x)	
10. POWER LEVEL 100			☐ 20.2203(a)(2)(ii)			☐ 50.36(c)(1)(ii)(A)			☐ 50.73(a)(2)(v)(A)		☐ 73.71(a)(4)	
			☐ 20.2203(a)(2)(iii)			☐ 50.36(c)(2)			☐ 50.73(a)(2)(v)(B)		☐ 73.71(a)(5)	
			☐ 20.2203(a)(2)(iv)			☐ 50.46(a)(3)(ii)			☒ 50.73(a)(2)(v)(C)		☐ 73.77(a)(1)	
			☐ 20.2203(a)(2)(v)			☐ 50.73(a)(2)(i)(A)			☒ 50.73(a)(2)(v)(D)		☐ 73.77(a)(2)(i)	
			☐ 20.2203(a)(2)(vi)			☐ 50.73(a)(2)(i)(B)			☐ 50.73(a)(2)(vii)		☐ 73.77(a)(2)(ii)	
						☐ 50.73(a)(2)(i)(C)			☐ OTHER		Specify in Abstract below or in NRC Form 366A	

12. LICENSEE CONTACT FOR THIS LER

LICENSEE CONTACT

Tracey Parmelee, Principal Engineer, Compliance

TELEPHONE NUMBER (Include Area Code)

(509) 377-8395

13. COMPLETE ONE LINE FOR EACH COMPONENT FAILURE DESCRIBED IN THIS REPORT

CAUSE	SYSTEM	COMPONENT	MANU-FACTURER	REPORTABLE TO EPIX	CAUSE	SYSTEM	COMPONENT	MANU-FACTURER	REPORTABLE TO EPIX
B	NG	V	BIF	Y					

14. SUPPLEMENTAL REPORT EXPECTED☐ YES (If yes, complete 15. EXPECTED SUBMISSION DATE) ☒ NO**15. EXPECTED SUBMISSION DATE**

MONTH	DAY	YEAR

ABSTRACT (Limit to 1400 spaces, i.e., approximately 15 single-spaced typewritten lines)

On October 3, 2016 at 1008 PDT, the Secondary Containment (Reactor Building) became inoperable due to pressure increasing above the Technical Specification limit of -0.25 inches of water gauge (inwg). While the plant was at 100% power, a Reactor Building exhaust valve (REA-V-1) unexpectedly closed, resulting in a loss of Secondary Containment vacuum for approximately four minutes. Operations personnel manually started the Standby Gas Treatment System A and quickly restored Secondary Containment to less than -0.25 inwg. While Technical Specification limits were exceeded for this short time period, the resulting pressure excursion was bounded by analytical results; and thus, there were no safety consequences for this condition. This event was reported under reporting criterion 10 CFR 50.72(b)(3)(v)(C) as Event Notification #52276.

The cause of the REA-V-1 closure was a loose termination point on the termination board, resulting in an intermittent loss of power to the valve. The corrective action for this event was to tighten the loose termination point screw.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

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1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER		
		YEAR	SEQUENTIAL NUMBER	REV NO.
Columbia Generating Station	05000- 397	2016	- 002	- 01

NARRATIVE**Plant Conditions**

At the time of the event, the plant was operating in Mode 1 at 100% power.

Event Description

On October 3, 2016 at 1008 PDT, Operations personnel received an annunciator alarm for high Secondary Containment [NH] differential pressure when the Reactor Building [NG] exhaust inboard isolation valve [V] (REA-V-1) failed shut. For a time period of approximately four minutes, Secondary Containment differential pressure increased above -0.25 inches of water gauge (inwg), resulting in Technical Specification (TS) 3.6.4.1 not being met.

This event is reportable as an event that could have prevented fulfillment of safety functions needed to control the release of radiation, and mitigate the consequences of an accident per 10 CFR 50.73(a)(2)(v)(C) and 10 CFR 50.73(a)(2)(v)(D). This condition was reported under 10 CFR 50.72(b)(3)(v)(C) via Event Notification #52276 for an event or condition that could have prevented fulfillment of a safety function needed to control the release of radioactive material.

Immediate Corrective Actions

Per plant procedure, Operations personnel manually started Standby Gas Treatment [BH] (SGT) System A and Secondary Containment was restored to less than -0.25 inwg at 1012 PDT.

Assessment of Safety Consequences

This event resulted in an unplanned entry into TS 3.6.4.1.A, in which Secondary Containment pressure was greater than -0.25 inwg for approximately four minutes. The peak pressure during this event was +4.17 inwg. While the actual pressure was beyond the range allowed by Technical Specifications, the purpose of maintaining a slight vacuum is to assist in drawdown of secondary containment to support accident response of the safety related SGT system. Existing engineering analysis demonstrates that for this event, the drawdown credited in Columbia's accident analysis could have been attained using either of the two available trains of the SGT system, thus there were no potential safety consequences. There was no actual safety consequence associated with this event since SGT did in fact restore Secondary Containment, and there was no loss of safety function or potential for radiological release.

Cause of Event

The apparent cause of the unexpected failure of REA-V-1 was a loose termination point on the termination board, resulting in an intermittent loss of power to the valve.



LICENSEE EVENT REPORT (LER) CONTINUATION SHEET

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1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER		
Columbia Generating Station	05000- 397	YEAR 2016	- SEQUENTIAL NUMBER 002	- REV NO. 01

NARRATIVE

Similar Events

A loss of the ability to maintain Secondary Containment pressure greater than required by Technical Specifications has occurred at Columbia Generating Station (Columbia) one time in the past two years, which was the result of a power supply failure. Prior to this two year time period, Columbia has experienced losses of Secondary Containment due to weather-related conditions, as well as human performance errors. Columbia has implemented corrective actions for these previous events.

Further Corrective Actions

An additional corrective action was to tighten the loose termination point screw. As no work has been performed on this equipment since original plant construction, the loosening of the screw is most likely due to vibration experienced from normal plant operation over a period of approximately 30 years; therefore, no further corrective actions are warranted.

Energy Industry Identification System (EIIS) Information codes from IEEE Standards 805-1984 and 803-1983 are represented in brackets as [XX] and [XXX] throughout the body of the narrative.